

IsDB-BISEW

Project Proposal

Student Management System (PHP & MySQL)

Trainee Name	**Md. Moheuzzaman**
Trainee ID	**1207808**
Batch	**PWAD/USSL-M/73/01**
Course Instructor	**Syed Ziaul Habib**
Course Consultant	**Md. Moshaidul Islam**

1. Project Overview

The Student Management System (SMS) is a web-based application developed using PHP and MySQL that enables educational institutions to efficiently manage student information, attendance, marks, fees, and academic records. It replaces manual record-keeping with a centralized, secure, and user-friendly digital platform.

2. Project Objectives

- Digitize and centralize student data management
- Reduce administrative workload and paperwork
- Provide quick, role-based access to student records
- Improve data accuracy and security
- Generate reports and analytics for decision-making

3. Scope of the Project

Module	Description
User Authentication	Login with role-based access (Admin, Teacher, Student)
Student Management	Add, edit, delete, and view student details
Course Management	Manage courses, subjects, and class sections
Attendance Management	Record and track daily attendance
Marks/Result Management	Enter marks and generate report cards
Teacher Management	Manage teacher profiles and subject assignments
Fee Management	Track fee payments and receipts
Reports & Analytics	Student lists, performance reports, summaries
Dashboard	Role-specific dashboards with statistics

4. Target Users

Administrator – Full access to all modules

Teacher – Manage attendance, marks, view student details

Student – View profile, attendance, and results

Parent (optional) – View child's academic progress

5. Technology Stack

Component	Technology
Frontend	HTML5, CSS3, JavaScript, Bootstrap
Backend	PHP (7.4+)
Database	MySQL
Server	Apache (XAMPP/WAMP/LAMP)
Version Control	Git (optional)

6. System Requirements

Hardware: Dual-core CPU, 4 GB RAM, 10 GB free disk space

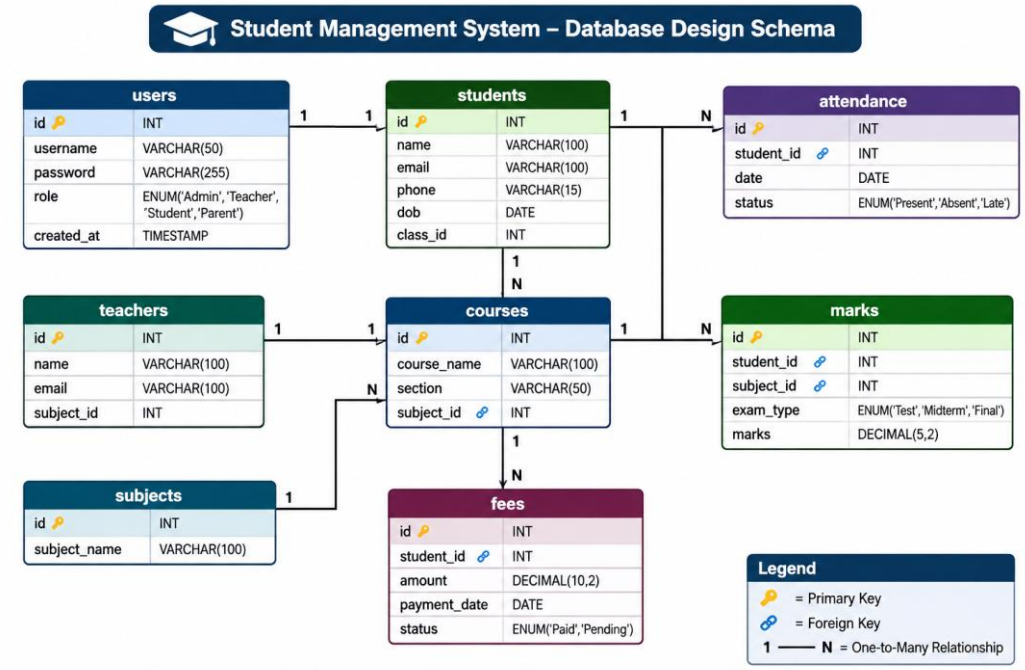
Software: XAMPP/WAMP/LAMP, MySQL Server, modern web browser

7. Database Design (Overview)

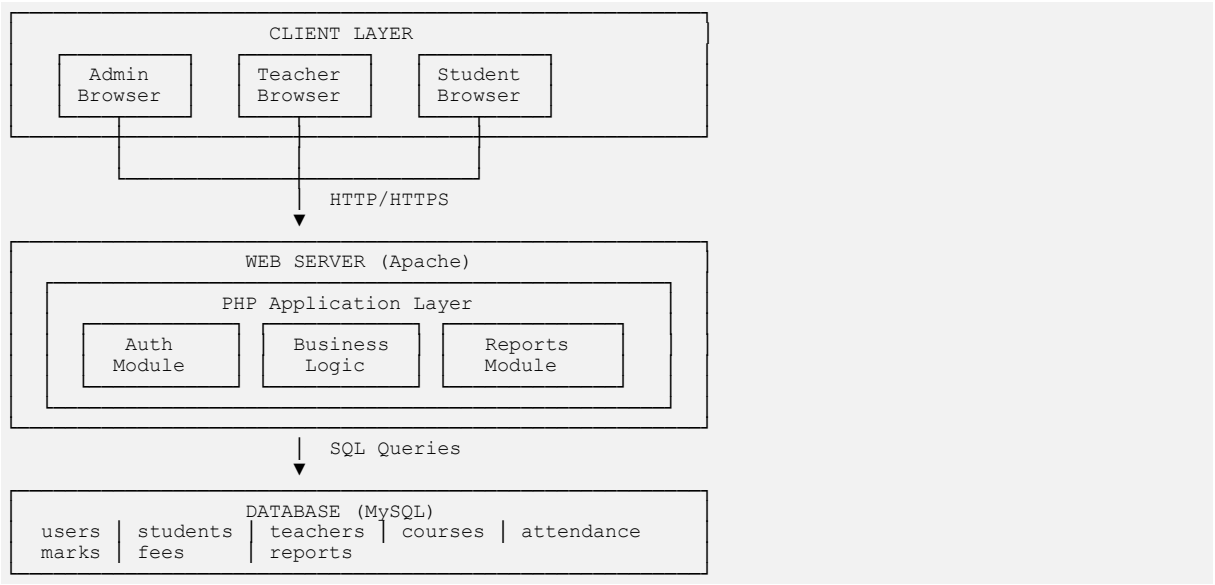
Key tables:

Table	Fields
users	id, username, password, role
students	id, name, email, phone, DOB, class_id
teachers	id, name, email, subject_id
courses	id, course_name, section
attendance	id, student_id, date, status
marks	id, student_id, subject_id, exam_type, marks
fees	id, student_id, amount, payment_date, status

Database Design Schema



8. System Architecture Diagram



9. Module Distribution

Module Workload Distribution

Module	% of Workload	Distribution
Student Management	25%	
Marks/Results	20%	
Attendance	15%	
Fees	15%	
Reports	15%	
Auth + Dashboard	10%	

10. Expected Benefits

Benefit Impact Analysis (Scale 1–10)

Benefit	Score	Impact (Scale 1–10)
Reduce manual work	10	
Faster report access	9	
Data accuracy	10	
Security	8	
Cost savings	7	
User satisfaction	8.5	

11. Risk Assessment (Matrix)

Likelihood ↓ / Impact →	Low	Medium	High
High		Data Loss	Server Downtime
Medium	Scope Creep	Security Breach	
Low			

Risk	Mitigation
Data loss	Regular backups
Security breach	Password hashing, prepared statements
Scope creep	Strict adherence to defined modules
Server downtime	Local dev server, cloud hosting for production

12. Conclusion

The Student Management System is a practical, scalable solution built on the proven PHP–MySQL stack. It offers a cost-effective, secure, and efficient way to manage student information while serving as an excellent demonstration of full-stack web development skills.